Date : 9th Sept 2020

Q Find integers x and y such that
$$gcd(272, 1479) = 272x + 1479y$$

Soln
$$1479 = 272 \times 5 + 119$$
$$272 = 119 \times 2 + 34$$
$$119 = 34 \times 3 + 17 \leftarrow \text{last non-zero remainder is gcd of } (272, 1479)$$
$$34 = 17 \times 2 + 0$$

$$gcd(272, 1479) = 17$$

So,

$$\Rightarrow 17 = ax + by$$
$$= 272x + 1479y$$

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Extended Euclidean Algorithm

Soln (cont)

$$\begin{aligned}
17 &= 119 - 34 \times 3 \\
&= 119 - [(272 - 119 \times 2) \times 3] \\
&= 119 - (272 \times 3 - 119 \times 6) \\
&= 119 - 272 \times 3 + 119 \times 6 \\
&= 119 \times 7 - 272 \times 3 \\
&= (1479 - 272 \times 5) \times 7 - 272 \times 3 \\
&= 1479 \times 7 - 272 \times 35 - 272 \times 3 \\
&= 1479 \times 7 - 272 \times 38 \\
17 &= 1479 \times 7 + 272 \times (-38)
\end{aligned}$$

$$\boxed{x = -38, \; y = 7}$$

Extended Euclidean Algorithm

$r_0 = a \qquad r_1 = b$

$s_0 = 1 \qquad s_1 = 0$

$t_0 = 0 \qquad t_1 = 1$

$\vdots \qquad \vdots$

$r_{i+1} = r_{i-1} - q_i \times r_i$

$s_{i+1} = s_{i-1} - q_i \times s_i$

$t_{i+1} = t_{i-1} - q_i \times t_i$

Computation stops when $r_{k+1} = 0$ and gives r_k as gcd of a and b

Ex- $a = 240, \; b = 46$

Soln

$240 = 5 \times 46 + 10$

$46 = 4 \times 10 + 6$

$10 = 1 \times 6 + 4$

$6 = 1 \times 4 + 2$

$4 = 2 \times 2 + 0$

So, $gcd(a,b) = 2$

$$gcd(a,b) = r_k = as_k + bt_k$$

i	q_{i-1}	r_i	s_i	t_i
0	—	240	1	0
1	—	46	0	1
2	$240 \div 46 = 5$	$240 - 5 \times 46 = 10$	$1 - 5 \times 0 = 1$	$0 - 0 \times 1 = 0$
3	$46 \div 10 = 4$	$46 - 4 \times 10 = 6$	$0 - 4 \times 1 = -4$	$1 - 4 \times (-5) = 21$
4	$10 \div 6 = 1$	$10 - 1 \times 6 = 4$	$1 - 1 \times (-4) = 5$	$-5 - 1 \times 21 = -26$
5	$6 \div 4 = 1$	$6 - 1 \times 4 = 2$	$-4 - 1 \times 5 = -9$	$21 - 1 \times (-26) = 47$
6	$4 \div 2 = 2$	$4 - 2 \times 2 = 0$	$5 - 2 \times (-9) = 23$	$-26 - 2 \times 47 = -120$

$r_k = as_k + bt_k$

$\Rightarrow -9 \times 240 + 47 \times 46 = 2$

(s_k, t_k)

$s = -9$

$t = 47$